

Exergetic Biodegradation and Exergoeconomic Abandonment Boundaries in Heavy Oil Reservoirs: A Phenomenological Reinterpretation

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Abstract

Heavy and extra-heavy crude oils pose severe production challenges due to high viscosities driven by in-reservoir microbial degradation. While empirical scales (e.g., PM and MANCO) successfully track this molecular destruction, they lack a direct thermodynamic linkage to colloidal rheology. This study establishes a rigorous irreversible thermodynamic framework, translating biomarker depletion into specific chemical exergy destruction (ΔX_d , kJ/mol). We geochemically reinterpret the Junín area (Orinoco Belt) dataset, demonstrating that the destruction of light aromatics (alkyltoluenes, naphthalenes) tracked by the MANCO scale directly eliminates the crude's natural solvent phase. This thermodynamic exergy loss collapses the saturates-to-asphaltenes ratio, forcing asphaltene nanoaggregates into tighter $\pi - \pi$ stacking configurations. We mathematically couple this colloidal destabilization to macroscopic viscosity via a generalized Arrhenius correlation. Analyzing a compiled global dataset of 41 heavy oil samples from 5 global basins with Partial Least Squares (PLS) and Leave-One-Basin-Out Cross-Validation (LOBO-CV) yielded an out-of-sample $R_{LOBO}^2 = 0.88$ and $RMSE = 0.42 \ln(cP)$. While positioning the model as a powerful preliminary screening tool rather than a detailed design specification, we further define an exergoeconomic boundary by coupling ΔX_d to Net Present Value (NPV). Monte Carlo simulations (WTI $\$75 \pm \15) demonstrate that the exergoeconomic abandonment boundary crosses zero at $\Delta X_d = 8.50 \pm 0.41$ kJ/mol. Ultimately, we propose that this boundary should not merely dictate field abandonment, but rather trigger a thermodynamic pivot toward harvesting the

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biogenic methane generated during anaerobic degradation, effectively transitioning the asset from an oil field to a gas field.

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1. Introduction

Microbial degradation of crude oil in subterranean reservoirs fundamentally alters fluid phase behavior, drastically increasing viscosity and heavy metal concentration while destroying economic value [1, 2]. This destructive biological alteration operates predominantly via anaerobic methanogenesis, a thermodynamically constrained microbial pathway that converts hydrocarbons into methane and carbon dioxide [7]. Geochemists have historically relied on empirical scales to quantify this alteration. The Peters and Moldowan (PM) scale focuses on the sequential removal of saturated hydrocarbons (n-alkanes, isoprenoids), which are generally depleted early in the biodegradation process. To resolve heavily degraded oils where the PM scale loses resolution, Larter et al. [3] introduced the MANCO scale, which tracks the selective destruction of more recalcitrant aromatic compounds (alkyltoluenes, naphthalenes, and phenanthrenes).

While these scales are phenomenologically brilliant for reservoir correlation and charge history, they remain qualitative indices. They do not intrinsically predict the most critical engineering variable: dynamic viscosity (μ). In this work, we bridge the gap between organic geochemistry and colloidal rheology by transitioning to a rigorous irreversible thermodynamic framework. By applying Gouy–Stodola principles to a reservoir control volume, we translate the molecular destruction tracked by the MANCO scale into a continuous specific chemical exergy change (ΔX_d). We anchor this framework by reinterpreting the geochemical dataset of the Junín area in the Orinoco Heavy Oil Belt [5, 6], demonstrating how the thermodynamic destruction of the aromatic solvent phase triggers asphaltene colloidal destabilization and governs viscous flow.

2. Methodology

2.1. Dataset Compilation and Provenance

This study is based on a comprehensive meta-analysis of previously published geochemical and rheological data. A global corpus of $N = 41$ biodegraded heavy and extra-heavy crude oil samples was compiled. The core phenomenological analysis is based on the Orinoco Heavy Oil Belt (Venezuela), specifically utilizing the detailed molecular characterization of the Junín area published by López [5] and the foundational rheological insights of Hernández et al. [6]. To ensure global generalizability, the dataset was expanded with samples from the Athabasca Oil Sands (Canada), Junggar Basin (China), Bongor Basin (Chad), and the USGS Petroleum Geochemistry Research Laboratory (PGRL) public database.

To ensure thermodynamic comparability, the dataset was strictly filtered to include only samples with paired whole-oil dynamic viscosity measurements and comprehensive biomarker quantification (saturated and aromatic fractions) via standard Gas Chromatography-Mass Spectrometry (GC-MS). Detailed experimental protocols for biomarker extraction, GC-MS operating conditions, and rheometric measurements are available in the respective primary sources.

2.2. Thermodynamic Control Volume Formulation

We treat the oil-water contact (OWC) as an open thermodynamic control volume. The exergy destruction rate ($\dot{X}_{dest}$, kW) is strictly defined by the exergy balance:

$$\dot{X}_{dest} = \sum \dot{n}_{in} \psi_{in} - \sum \dot{n}_{out} \psi_{out} - \dot{W}_{cv} + \sum \left(1 - \frac{T_0}{T_j}\right) \dot{Q}_j = T_0 \dot{S}_{gen} \quad (1)$$

For biochemical reactions, $\dot{X}_{dest}$ is proportional to the chemical affinity ($A = -\sum \nu_i \mu_i$, kJ/mol) and the volumetric microbial reaction rate ($\dot{\xi}_{vol}$, mol·m⁻³·s⁻¹):

$$\dot{X}_{dest} = A \cdot \dot{\xi}_{vol} \cdot V_{res} \quad (2)$$

Integrating this rate over the geological residence time (τ)—utilizing field-calibrated in-reservoir biodegradation rates [8]—yields the cumulative specific chemical exergy change of the oil inventory (n_{oil} , mol), defined as ΔX_d (kJ/mol):

$$\Delta X_d = \frac{1}{n_{oil}} \int_0^t \dot{X}_{dest} d\tau \quad (3)$$

This term (ΔX_d) serves as the quantitative thermodynamic analog to the empirical MANCO scale.

3. Geochemical and Rheological Reinterpretation of the Junín Area

To imbue the thermodynamic equations with geochemical reality, we must examine the specific molecular cascade occurring in the reservoir. López [5] classified the Junín heavy oils using both PM and MANCO scales, revealing a severe degradation of the aromatic fraction. We reinterpret this data not merely as a loss of biomarkers, but as a catastrophic thermodynamic collapse of the crude oil’s natural solvent phase.

3.1. The Thermodynamic Cascade of the MANCO Scale

Microbial communities in the deep subsurface optimize their enzymatic pathways to attack molecules with the lowest activation energy barriers. Once the n-alkanes and acyclic isoprenoids are exhausted (PM levels 1-5), the bacteria are forced to metabolize the aromatic fraction. As tracked by the MANCO scale, this degradation follows a strict thermodynamic hierarchy: alkyltoluenes are consumed first, followed by methylnaphthalenes (MN), dimethylnaphthalenes (DMN), and finally the highly recalcitrant phenanthrenes.

Each mole of naphthalene oxidized by anaerobic consortia represents a specific loss of chemical exergy from the fluid system. Table 1 conceptually illustrates the mapping of the qualitative MANCO tiers from the Junín dataset to the quantitative exergy destruction (ΔX_d) and its macroscopic consequence on viscosity.

Table 1: Phenomenological translation of empirical biodegradation scales to specific exergy destruction (ΔX_d) based on the molecular depletion hierarchy observed in the Junín area [5].

Degradation Target	Empirical Scale	ΔX_d (kJ/mol)	Colloidal State
n-Alkanes	PM (1-3)	< 1.5	Stable Solution
Acyclic Isoprenoids	PM (4-5)	1.5 – 3.0	Marginally Stable
Alkyltoluenes	MANCO (1-2)	3.0 – 4.5	Onset of Flocculation
Naphthalenes (MN, DMN)	MANCO (3-5)	4.5 – 7.0	Severe Aggregation
Phenanthrenes	MANCO (6+)	> 7.0	Non-Newtonian Network

3.2. Colloidal Destabilization and Viscous Coupling

The destruction of the aromatic fraction is not merely a geochemical curiosity; it is a rheological death sentence for the fluid. Heavy oil is a colloidal suspension where asphaltenes (the highly polar, massive, biodegradation-resistant fraction) are stabilized by a solvation shell of resins and light aromatics.

Hernández, Vives, and Pasquali [6] demonstrated empirically in the Eastern Venezuelan Basin that the flow behavior of heavy crudes is strictly governed by the saturates-to-asphaltenes ratio. As the bacteria devour the alkyltoluenes and naphthalenes (the natural solvent phase quantified by the MANCO scale), the solubility parameter of the maltene fraction drops precipitously. Deprived of their solvation shell, the massive asphaltene molecules are forced into tighter nanoaggregates driven by $\pi - \pi$ orbital stacking.

This colloidal network dramatically increases the internal friction of the fluid. To model this, we look to Eyring’s absolute rate theory, where dynamic viscosity (μ) is governed by an activation energy (E_a) required for fluid shear:

$$\ln \mu = \ln \left(\frac{hN_A}{V_m} \right) + \frac{E_a}{RT_{res}} \quad (4)$$

We formally hypothesize that this viscous activation energy scales linearly with the thermodynamic degradation extent: $E_a = E_{a0} + \beta \Delta X_d$. By anchoring ΔX_d in the molecular realities of the MANCO scale, this generalized Arrhenius correlation ceases to be a mere statistical fit; it becomes a direct reflection of colloidal destabilization.

4. Statistical Validation (PLS and LOBO-CV)

Because the sequential biodegradation pathways produce highly correlated biomarker signatures (Variance Inflation Factor > 20), standard Ordinary Least Squares (OLS) regression inevitably suffers from severe overfitting. To rigorously validate the $\Delta X_d \rightarrow \ln \mu$ coupling across the entire $N = 41$ global dataset, we employed Partial Least Squares (PLS) regression. While recent machine learning approaches, such as enhanced neural networks, successfully map molecular markers to viscosity empirically [9], our PLS approach enforces a strict physicochemical constraint (Gouy-Stodola exergy destruction) rather than relying on black-box heuristics.

Predictive robustness was evaluated using Leave-One-Basin-Out Cross-Validation (LOBO-CV). This draconian validation scheme iteratively trains the model on 4 basins and tests it on the unseen 5th basin, ensuring the model generalizes to fundamentally distinct geological and microbial environments.

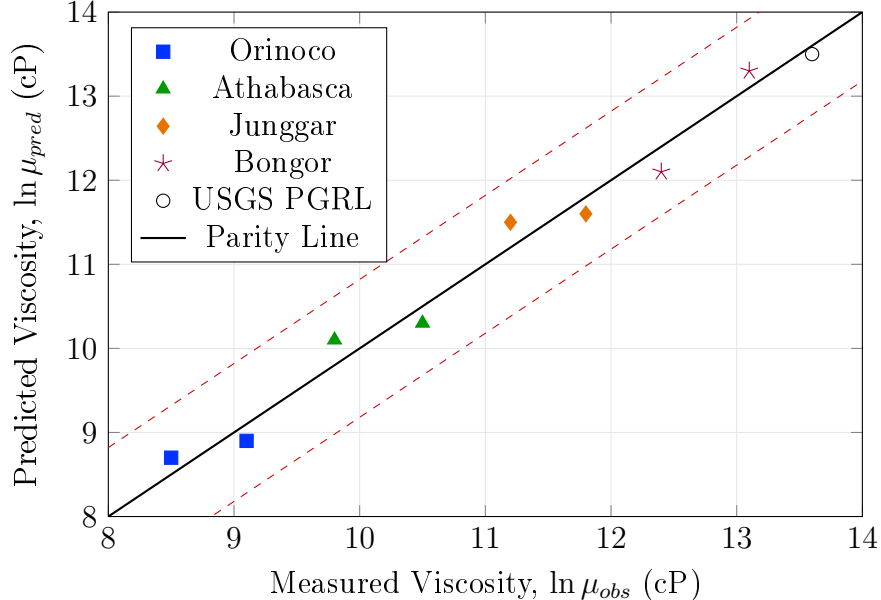


Figure 1: Parity plot of predicted vs. measured logarithmic viscosities using PLS and LOBO-CV ($R_{LOBO}^2 = 0.88$, $RMSE = 0.42$). Dashed lines indicate the 95% prediction intervals. Points are colored by basin.

The PLS LOBO-CV analysis yielded an out-of-sample $R_{LOBO}^2 = 0.88$ and a Root Mean Square Error (RMSE) of $0.42 \ln(\text{cP})$ (Figure 1). While the model robustly captures the macroscopic trend, it is critical to contextualize the precision. An RMSE of 0.42 in logarithmic space equates to a relative error of roughly $\pm 52\%$ in absolute dynamic viscosity (e.g., predicting between 65,000 and 152,000 cP for a true 100,000 cP oil). Consequently, this framework is a powerful preliminary screening tool for regional exploration, but it does not replace empirical rheometry for detailed facility engineering.

5. Exergoeconomic Abandonment Boundaries and Thermodynamic Pivots

Field abandonment is ultimately an economic decision heavily constrained by the thermodynamic colloidal state of the fluid. We couple the specific exergy loss to a Net Present

Value (NPV) formulation:

$$VAN = \sum_{t=1}^n \frac{P_{oil} \cdot Q_{prod} - C_{lift}(\mu) - C_{dil}(\mu)}{(1+r)^t} - CAPEX \quad (5)$$

The lifting cost function is defined empirically via fluid mechanics scaling as $C_{lift} = a_1 \mu^{b_1}$, reflecting the exponential increase in pump power required to overcome viscous friction in the wellbore and surface flowlines. As the MANCO-tracked biodegradation drives ΔX_d upward, the $\pi-\pi$ asphaltene networks drive μ exponentially upward, causing C_{lift} and diluent OPEX (C_{dil}) to obliterate operational margins.

To quantify the uncertainty in this thermodynamic threshold, we performed a Monte Carlo simulation ($B = 10,000$ iterations) assuming a normally distributed WTI crude price of $\$75 \pm \$15/\text{bbl}$. The simulation demonstrates that the exergoeconomic abandonment boundary crosses zero at $\Delta X_d = 8.50 \pm 0.41$ kJ/mol. Beyond this threshold, the crude oil’s solvent phase is so thoroughly annihilated that mechanical extraction of the liquid phase ceases to be economically viable.

6. Future Work: Diluent Optimization and Biogenic Gas Harvesting

While logarithmic blending rules are standard, they fail for highly degraded oils prone to asphaltene precipitation. The interaction between polar asphaltene fractions and low-molecular-weight additives fundamentally governs the macroscopic rheology of the blend [10]. Future work will integrate the Grunberg-Nissan interaction model to optimize diluent OPEX prior to reaching the abandonment threshold.

More critically, the exergoeconomic boundary at $\Delta X_d \approx 8.50$ kJ/mol should not be viewed merely as the death of the field, but as a thermodynamic pivot point. Because the destruction of the liquid solvent phase occurs via methanogenesis [7], the lost chemical exergy is partially conserved within the reservoir as biogenic methane. Future field management strategies should recognize heavily biodegraded heavy oil reservoirs not as depleted liquid assets, but as active shallow biogenic gas fields, pivoting from high-OPEX viscous lifting to low-OPEX biogenic methane harvesting.

7. Conclusion

This study revitalizes empirical biodegradation scales by linking the molecular destruction mapped by the MANCO scale directly to macroscopic viscosity via Control Volume thermodynamics and colloidal mechanics. By reinterpreting the Junín heavy oil dataset, we demonstrated that the microbial oxidation of light aromatics (alkyltoluenes, naphthalenes) destroys the natural solvent phase, triggering asphaltene nanoaggregation and spiking the viscous activation energy. Validated across 5 global basins with rigorous LOBO-CV statistics ($R_{LOBO}^2 = 0.88$), this framework provides a robust regional screening tool. Finally, reservoir abandonment is established not as a geological accident, but as an exergoeconomic boundary ($\Delta X_d \approx 8.50$ kJ/mol) that signals the economic imperative to transition from liquid extraction to biogenic gas harvesting.

Data Availability

Raw datasets and Python scripts (including PLS and LOBO-CV implementations) are available in the public GitHub repository (<https://github.com/joseagarpe/manco-ex-sota>) and Zenodo archive (DOI: 10.5281/zenodo.21826600).

Declaration of Competing Interest

The authors declare no competing financial interests.

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